

FleetPath — Fuel Pathway Decision Report

Independent cost and carbon analysis for fleet procurement decisions

Fleet Profile

REGION	VEHICLE CLASS	FLEET SIZE	ANNUAL MILEAGE	LIFECYCLE HORIZON
TX	school_bus_typeC	25 vehicles	15,000 mi	10 yr

TL;DR Verdict

Battery-Electric wins: no capital payback within fleet lifecycle versus the diesel baseline, cutting lifecycle CO2e emissions by 60.5% relative to diesel. Note: Battery-Electric's lifetime cost is \$598,972 USD higher than diesel's over the fleet lifecycle -- this pathway is favored on blended cost/carbon utility, not on lowest raw cost.

Payback vs. diesel: N/A | Emissions reduction vs. diesel: 60.5%

Pathway Comparison

Fuel Pathway	Lifetime TCO	Lifecycle CO2e	Cold-Climate Adj.
Diesel	\$675,000 USD	635.0 t	No
Battery-Electric (Winner)	\$1,273,972 USD	250.6 t	No
Hydrogen Fuel Cell	\$3,119,375 USD	563.8 t	No
CNG	\$733,750 USD	423.7 t	No
Biodiesel (B20)	\$679,078 USD	518.9 t	No

Methodology: figures derived from NREL H2A hydrogen station cost models, Argonne National Laboratory AFLEET 2023 vehicle and infrastructure baselines, EPA eGRID 2023 Rev 2 regional grid emission factors (US), and Environment and Climate Change Canada National Inventory Report 2023 (NIR 1990-2021) provincial grid intensities (Canada). This is a decision-support estimate, not a procurement guarantee.